Conference Presentation Slide Content (15 min)

On-slide bullets and figures for each slide. Bullets are written tight for projection (keep spoken detail in your notes, not on the slide). Figure files referenced live in this same folder.

Slide 1 - Title

Title: Strategies for Hybrid, Hands-on Courses in Robotics, Embedded Systems, and IoT

- Jonathan Jaramillo, Cornell University, Systems Engineering Program
- Conference name and date
- Tag line: A Work-in-Progress case study

Figure: none (optional institution logo)

Slide 2 - The Problem

- Online enrollment doubled: 18% to 45% exclusively-online (2019 to 2020)
- Structural shift, not just COVID: professional M.Eng. growth, working professionals, employer-funded part-time degrees
- Hands-on hardware labs are the hardest experience to deliver remotely
- They depend on physical manipulation and sensory feedback

Figure: none, or a simple enrollment-trend visual

Slide 3 - Why Existing Solutions Fall Short

- Simulations and remote-access labs: keep students at arm's length, "black box" effect
- Take-home kits: preserve real debugging, but new costs
    - Lost peer coordination
    - Weaker oral communication
    - Harder integrity verification
- New wrinkle: generative AI can produce code and reports without understanding

Figure: none

Slide 4 - Research Questions

1. Do standardized at-home kits achieve technical and pedagogical parity?
2. Do periodic oral assessments support engagement and integrity in the age of AI?
3. Does the flipped format increase perceived parity between tracks?

- Questions emerged from building the courses; this paper lays groundwork

Figure: none

Slide 5 - The Two Courses

- SYSEN 5411 - Introduction to Robotics
- SYSEN 5412 - Creating Solutions with Embedded Systems
- Both one-credit, hybrid, flipped; identical kits shipped to remote students
- Cohort: 44 students total
    - 27 on-campus (full-time, one-year M.Eng.)
    - 17 distance learners (working professionals, multi-year part-time)

Figure: major_distribution_pie_chart.png (cohort composition)

Slide 6 - Design Decision 1: Flipped + Hybrid

- Lectures delivered asynchronously (pre-recorded video)
- Weekly synchronous sessions reserved for demos and troubleshooting
- Sessions recorded for remote students
- Ed Discussion peer support, incentivized with participation points
- Payoff: synchronous time freed for the hands-on work students valued most

Figure: none, or a simple flipped-model schematic

Slide 7 - Design Decision 2: Hardware + Language

- Parity-driven hardware: no soldering iron or oscilloscope, strong online docs
    - Robotics: XRP (Experiential Robotics Platform), web-based IDE
    - Embedded: SunFounder Raspberry Pi Pico Starter Kit
- Language: MicroPython
    - 93% of students already knew Python (41 of 44)
    - Far fewer knew C (12) or C++ (20); 43% were hardware novices
- Goal: cut the syntax and tooling hurdle, spend cognitive load on systems logic

Figure: programming_languages_pie_chart.png

Slide 8 - Assessment: Oral Exams + Workload

- Oral exam: 20-minute Zoom design review after the planning phase
    - User interaction, system mapping (use cases to requirements to state machine), implementation strategy (e.g. polling vs. interrupts)

- Dual purpose: AI-mitigation tool and relational touchpoint
- Workload reality:
  - Robotics spread up to 7 to 9 hours/week
  - Root cause of outliers: battery brown-outs, fixable in seconds
  - Response: "two-hour rule" and robotics final project made optional

Figure: none (the battery story is spoken, not on-slide)

Slide 9 - Early Result: The Engagement Surprise

- Expectation: distance learners would disengage
- Reality: distance learners were more engaged and more ambitious
  - Treated the project as a creative break from work
  - On-campus students, overloaded, defaulted to minimum viable products
- Oral exams reliably surfaced skipped foundational labs
  - Near-perfect correlation: incomplete weekly work to weak design review

Figure: none

Slide 10 - Early Result: AI Usage (Lab 1)

- ~36% used no AI at all
- Of reported uses, debugging dominated (~30%)
- Conceptual and code-writing uses were rare
- Read: AI used to clear friction points, not to replace learning
- Reinforces oral exams: AI handles the "how," students explain the "why"

Figure: ai_usage_pie_chart.png

Slide 11 - Future Work

- More prewritten boilerplate to shift robotics focus to systems integration
- Tiered oral exam schedule: 3 checkpoints (Weeks 1-4, 5-8, final review)
- Systematically log and categorize hardware vs. software friction points
- Classify AI use as "augmentative" vs. "bypass," correlate with oral exam performance
- Seek IRB approval for demographic analysis (gender, industry background)

Figure: none

Slide 12 - Conclusion

- Parity needs pedagogy, not just identical hardware

- Tool-less kits cut logistical friction
- Oral exams turn assessment into authentic mentorship, not an AI-vulnerable task
- Foundation for a longitudinal study on hybrid hardware instruction

Figure: none

Slide 13 - Questions / Contact

- Thank you
- Name, email, course or project links
- Invitation for questions

Figure: none. Backup slide: familiarity ranking chart (Figure 3 from paper) for detailed questions

Figure Inventory

- major_distribution_pie_chart.png - Slide 5
- programming_languages_pie_chart.png - Slide 7
- ai_usage_pie_chart.png - Slide 10
- Familiarity ranking (paper Figure 3) - backup slide only
- Three figures in the deck keeps it under chart-fatigue threshold